DAC 2027: compiled results

Our accelerator vs PTB, SpikeX, LoAS, SATO, SpikeFlow, Prosperity (and Phi for storage). Generated from the result files of `dac2027_sim`; every number below is read from those files.

1. Setup

- **Workload *sparse90*** (all main results): 90% of int8 weights zero; 90% of neurons silent; a non-silent neuron fires with p = 0.3 per timestep and at least once; T = 8 (about 3.2% of spike bits are 1). VGG16 and ResNet18 (CIFAR shapes).
- **OP** = one accumulation per '1' spike bit per output channel (Phi's definition): VGG16 65.4 M OP/image, ResNet18 117.7 M OP/image, the same for every design.
- **Arrays**: every design at the size given in its own paper (table below). Common clock 500 MHz (the clock our pipelined core closes at 45 nm); normalized speed-ups do not depend on the clock.
- **Technology**: all designs synthesized from our RTL of each design at 45 nm (Nangate45, Yosys + ABC), gate-level power with OpenSTA; SRAM with CACTI 7 at 45 nm (LSTP cells, LOP periphery, 300 K; every buffer meets 500 MHz); DRAM 12.45 pJ/bit (Prosperity repo, from DRAMsim3).

Design	Array (from its paper)	Cycle numbers from
PTB	16 x 8 PEs (128 adders)	official Prosperity-repo PTB model
SpikeX	8 x 8 PEs (64)	our cycle model of the paper
LoAS	16 TPPEs	our cycle model of the paper (loas_model.py)
SATO	128 PEs	official Prosperity-repo SATO model
SpikeFlow	16 x 16 PEs (256)	our cycle model of the paper
Prosperity	128 PEs	official Prosperity simulator
Ours	4 x 4 PEs x 8 adders (128)	cycle model, = RTL cycle-exact

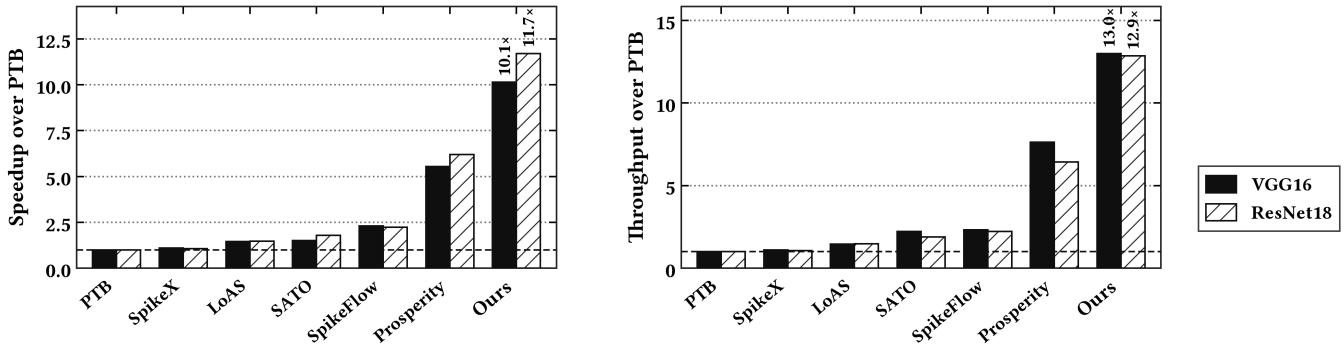
2. RTL verification of our design

Closed-loop Verilator testbench: the RTL packer forms the groups, the output SRAM is synchronous, the controller follows the cycle model. Every layer is checked for partial sums, output spikes and cycle count.

Network	Batch	Layers OK	RTL cycles	Model cycles	Sum/spike mismatches
VGG16	1	16/16	364,930	364,930	0
VGG16	16	16/16	4,576,997	4,576,997	0
ResNet18	1	21/21	543,056	543,056	0
ResNet18	16	21/21	7,977,322	7,977,322	0

3. Latency and throughput (paper array sizes)

Network	Design	Cycles (batch 1)	Latency @500 MHz (ms)	Latency vs PTB	Throughput, batch 16 (img/s)	Throughput vs PTB
VGG16	PTB	3,701,483	7.403	1.000	135	1.00x
VGG16	SpikeX	3,335,686	6.671	0.901	150	1.11x
VGG16	LoAS	2,548,501	5.097	0.689	197	1.46x
VGG16	SATO	2,448,376	4.897	0.661	299	2.22x
VGG16	SpikeFlow	1,601,259	3.203	0.433	314	2.33x
VGG16	Prosperity	666,778	1.334	0.180	1,026	7.62x
VGG16	**Ours**	364,930	0.730	0.099	1,748	**12.99x**
ResNet18	PTB	6,352,976	12.706	1.000	78	1.00x
ResNet18	SpikeX	5,977,620	11.955	0.941	83	1.06x
ResNet18	LoAS	4,336,311	8.673	0.683	115	1.48x
ResNet18	SATO	3,555,140	7.110	0.560	147	1.88x
ResNet18	SpikeFlow	2,854,289	5.709	0.449	174	2.23x
ResNet18	Prosperity	1,025,513	2.051	0.161	502	6.44x
ResNet18	**Ours**	543,056	1.086	0.085	1,003	**12.85x**



(a) Latency (batch 1)

(b) Throughput (batch 16)

Fig. 1. (a) Speedup (latency, batch 1) and (b) throughput (batch 16) over PTB, sparse90, paper array sizes.

Dense-weight workload *zipstats* (weights dense, 26.1% non-silent neurons): where our weight packing has nothing to skip.

Network	Design	Cycles (batch 1)	Latency vs ours (>1 = slower than ours)	Throughput vs ours (>1 = faster than ours)
VGG16	PTB	7,439,307	1.62	0.49
VGG16	SpikeX	8,658,454	1.88	0.42
VGG16	LoAS	7,498,876	1.63	0.49
VGG16	SATO	4,541,498	0.99	1.45
VGG16	SpikeFlow	3,869,059	0.84	0.95
VGG16	Prosperity	1,117,161	0.24	3.68
VGG16	**Ours**	4,602,491	1.00	1.00
ResNet18	PTB	12,031,392	1.77	0.53
ResNet18	SpikeX	15,711,996	2.31	0.40
ResNet18	LoAS	13,196,528	1.94	0.48
ResNet18	SATO	5,254,034	0.77	1.30
ResNet18	SpikeFlow	6,912,849	1.02	0.92
ResNet18	Prosperity	2,034,105	0.30	3.19
ResNet18	**Ours**	6,796,249	1.00	1.00

4. Packing

Measured on sparse90, batch 1. Filter packing: filters with a non-zero weight in a K-tile share a column when their non-zeros never fall in the same row (up to F = 4). Pixel packing: pixels whose spike bits never collide in a K-tile share one group (up to G = 4); pixels with no spike in the tile are skipped.

Network	Filters per column	Non-zero filter slots / columns	Pixels per group	Non-zero pixel steps / groups
VGG16	2.46	2,888,812 / 1,172,311	3.44	128,975 / 37,478
ResNet18	2.47	959,197 / 388,969	3.57	337,316 / 94,455

Storage compression = dense stored bits / packed stored bits (weights + spikes), 128 x 128 matrices, weight sparsity = neuron sparsity = x; higher is better. Each design's own format.

Sparsity	Ours	LoAS	PTB	Phi	Prosperity
50%	0.71	1.58	0.94	1.00	1.00
70%	1.19	2.30	1.13	1.19	1.00
80%	1.69	2.99	1.25	1.32	1.00
90%	3.00	4.26	1.40	1.50	1.00
95%	5.41	5.41	1.49	1.64	1.00
98%	10.88	6.46	1.55	1.77	1.00

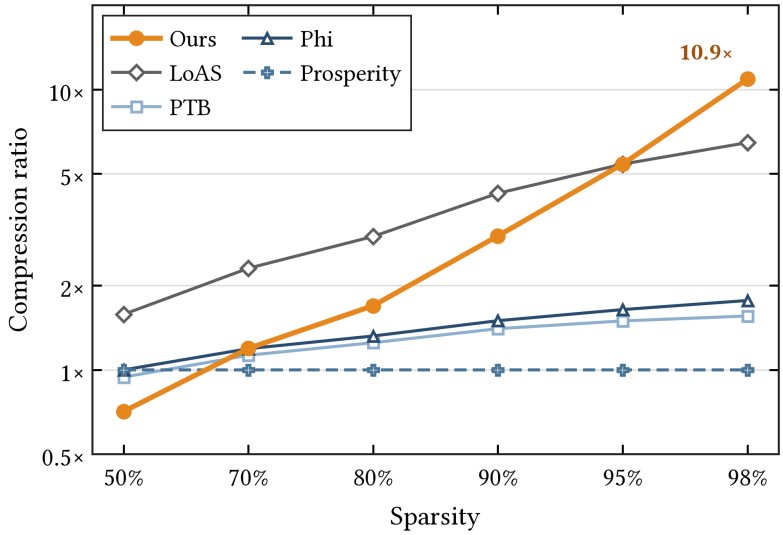


Fig. 2. Storage compression ratio vs sparsity.

Validation of the storage models against the papers' own numbers:

Check	Paper	This model
Prosperity product sparsity, VGG16 CIFAR100 real spikes (official simulator)	bit 34.21% → 2.79% (Table I/II)	bit 34.21% → 2.79% (exact)
Prosperity, VGG16 CIFAR10 / ResNet18 CIFAR100	not reported	8.96% → 2.48% / 8.63% → 2.63%
Phi Table 4 random 5% (bit/L1/L2+/L2-, over bit)	5.0/2.4/2.6/0.0, 2.0x	5.0/2.7/2.4/0.1, 2.0x
Phi Table 4 random 10%	10.0/6.6/3.4/0.0, 2.9x	10.0/6.8/3.6/0.3, 2.6x
Phi Table 4 random 20%	19.9/13.9/6.4/0.4, 2.9x	20.0/15.4/6.2/1.6, 2.6x
Phi Table 4 random 50%	50.0/49.8/7.9/7.7, 3.2x	50.0/49.4/8.0/7.3, 3.3x
Phi on real VGG16 / ResNet18 CIFAR10 spikes (over bit)	5.8x / 3.7x on the authors' own models (8.7% / 7.4% bit density)	3.8x / 3.9x on Prosperity's models (9.0% / 9.8%)
Phi activation traffic vs dense (Fig 12a)	~0.47-0.52, geomean ~0.49	0.475 (VGG16), 0.466 (ResNet18)
Phi weight + PWP traffic without prefetcher (Fig 12b)	~9x dense weights	9x (8-bit PWPs)
PTB, LoAS, SpikeX, SpikeFlow	report no compression ratio that can be reproduced from their text	format only

5. Area and power breakdown of our core (45 nm)

Block	Area (mm²)	Share	Power @500 MHz (mW)	Share	Worst path (ns)
PE array (16 PEs)	0.2190	23.0%	89.7	30.3%	0.86
edge FIFOs (4 x 80 entries)	0.4804	50.6%	118.7	40.1%	1.96
packer (8 slices/cycle)	0.1056	11.1%	39.4	13.3%	1.30
LIF units (16)	0.0975	10.3%	26.2	8.8%	1.88
arbiter, skew, id sideband, RMW adders	0.0476	5.0%	22.3	7.5%	1.59
Core total	**0.9501**	100%	**296.3**	100%	**1.96**

Yosys + ABC (buffered), Nangate45 typical; power from gate-level activity of the closed-loop testbench on real VGG16 layer data; 0 unannotated pins. Synthesis-level timing (no placement): 1.96 ns worst path, so 500 MHz.

6. All designs at 45 nm, 500 MHz: area, GOP/s, GOP/J

Design	Core (mm²)	SRAM (mm²)	Total (mm²)	Core power (mW)	VGG16 GOP/s	GOP/J core	GOP/J core+SRAM	GOP/J system	ResNet18 GOP/s	GOP/J core	GOP/J core+SRAM	GOP/J system
PTB	0.953	0.279	1.232	381.4	8.8	23	22	10	9.2	24	24	19
SpikeX	0.468	0.279	0.747	187.8	9.8	52	50	32	9.7	52	50	39
LoAS	0.370	1.108	1.478	57.7	12.9	223	153	116	13.6	235	162	126
SATO	0.076	0.279	0.355	21.6	19.5	903	421	73	17.3	800	406	123
SpikeFlow	0.123	0.084	0.207	47.2	20.5	435	385	96	20.4	433	393	151
Prosperity	0.369	0.666	1.035	239.8	67.1	280	85	34	59.1	247	85	36
Ours	0.950	1.178	2.128	296.3	114.3	386	148	**105**	118.1	398	150	**115**

Per image at batch 16. System = core + SRAM (leakage and read/write energy from each design's access counts) + DRAM.

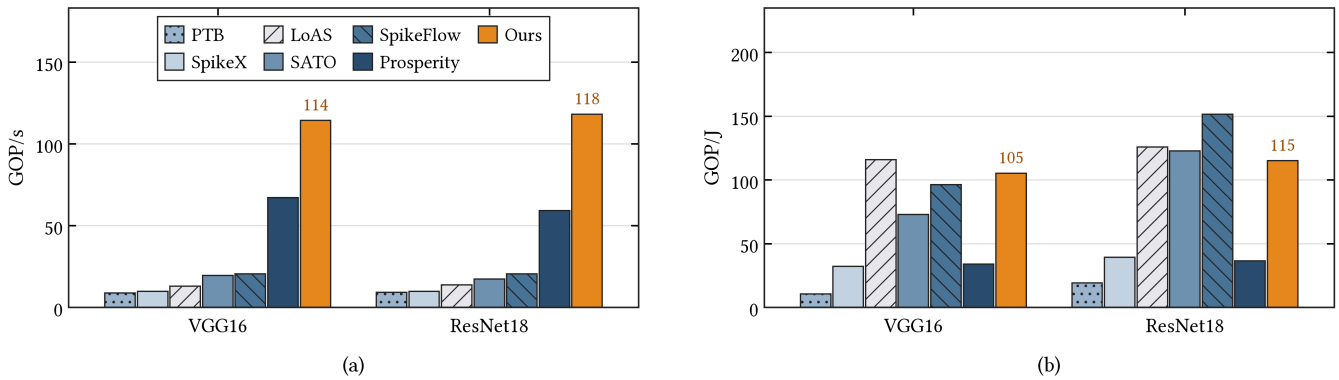


Fig. 3. (a) GOP/s and (b) system GOP/J at 45 nm, 500 MHz.

7. Power → energy → EDP (core / SRAM / DRAM), batch 1

Power is derived first for each component, averaged over one batch-1 inference of latency t: core = gate-level power; SRAM = leakage + (read/write energy from the access counts) / t; DRAM = 12.45 pJ/bit × DRAM traffic / t. Energy = power × latency; EDP = energy × latency.

Network	Design	Latency (ms)	Core (mW)	SRAM (mW)	DRAM (mW)	Total (mW)	Core (µJ)	SRAM (µJ)	DRAM (µJ)	Energy (µJ)	EDP (µJ·ms)
VGG16	PTB	7.403	381	10	455	847	2824	76	3372	6272	46434
VGG16	SpikeX	6.671	188	12	512	712	1253	79	3418	4750	31690
VGG16	LoAS	5.097	58	31	162	251	294	159	824	1278	6512
VGG16	SATO	4.897	22	24	698	744	106	118	3418	3642	17834
VGG16	SpikeFlow	3.203	47	21	1057	1126	151	68	3386	3605	11545
VGG16	Prosperity	1.334	240	426	2877	3542	320	568	3836	4724	6300
VGG16	**Ours**	0.730	296	395	1345	2036	216	288	982	**1486**	**1085**
ResNet18	PTB	12.706	381	6	93	481	4847	73	1186	6105	77573
ResNet18	SpikeX	11.955	188	7	113	308	2245	81	1355	3681	44007
ResNet18	LoAS	8.673	58	26	48	132	501	229	416	1145	9931
ResNet18	SATO	7.110	22	21	191	234	154	151	1355	1660	11805
ResNet18	SpikeFlow	5.709	47	7	212	266	269	40	1210	1519	8673
ResNet18	Prosperity	2.051	240	444	1106	1790	492	911	2268	3671	7529
ResNet18	**Ours**	1.086	296	456	403	1155	322	495	438	**1254**	**1362**

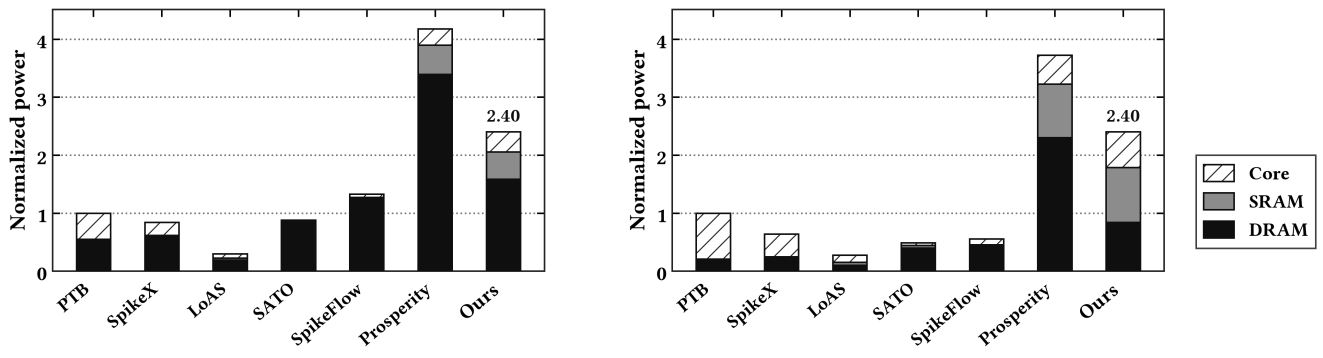
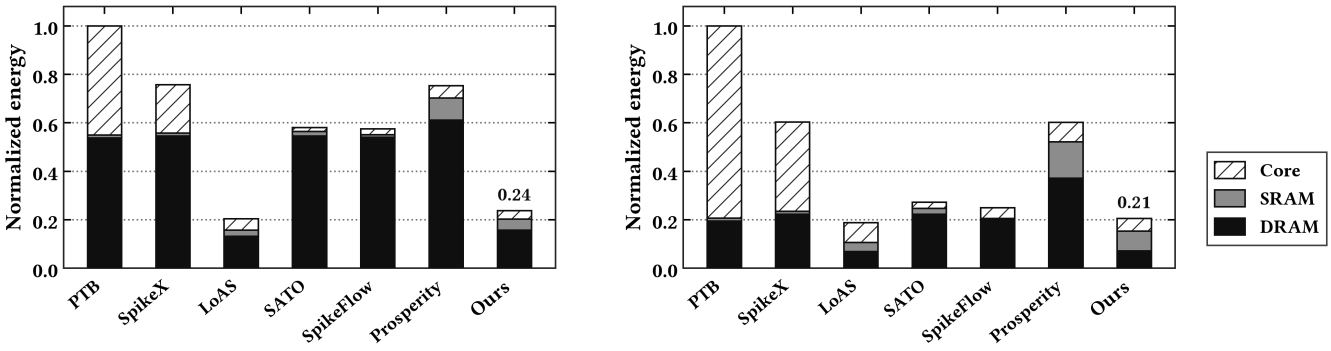
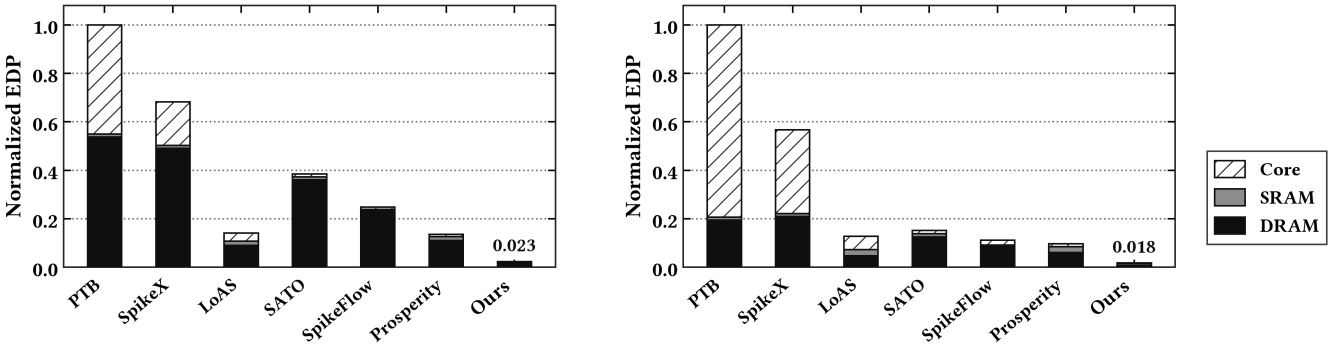


Fig. 4. Power per component (DRAM / SRAM / core), normalized to PTB = 1, batch 1.



(a) VGG16 (b) ResNet18

Fig. 5. Energy per image (power × latency) per component, normalized to PTB = 1, batch 1.



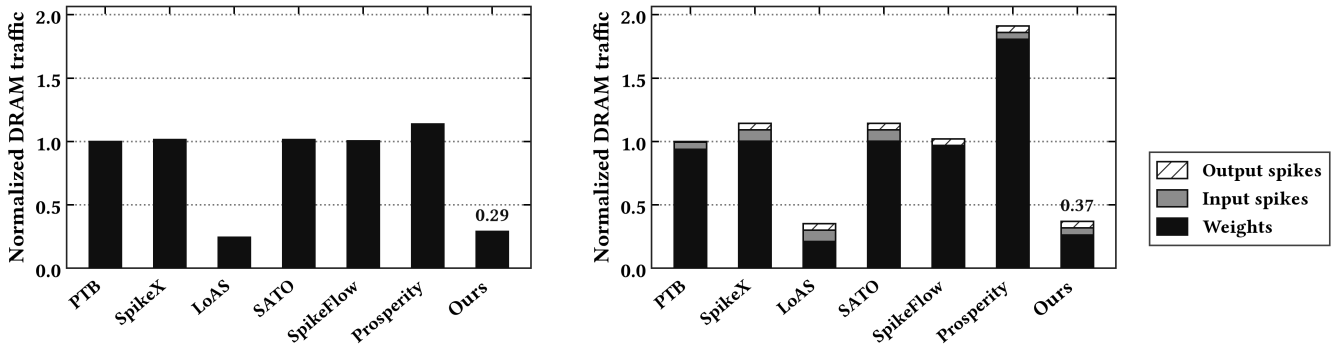
(a) VGG16 (b) ResNet18

Fig. 6. EDP (energy × latency) split by where the energy goes, normalized to PTB = 1, batch 1.

8. Memory traffic

DRAM traffic per image. Prosperity: official simulator (counters split into weights and activations); PTB: official model (one image at a time); others: the same tiled rule with each design's own format and buffer. Ours stores weights as packed columns (8 B per column) and reads raw spike trains.

Network	Design	Weights b1 (MB)	Input spikes b1	Output spikes b1	Total b1	Total b16 (per image)
VGG16	PTB	33.63	0.194	0.036	33.86	33.86
VGG16	SpikeX	33.77	0.260	0.285	34.32	7.45
VGG16	LoAS	7.57	0.421	0.285	8.27	1.39
VGG16	SATO	33.77	0.260	0.285	34.32	7.45
VGG16	SpikeFlow	33.66	0.044	0.285	33.99	5.12
VGG16	Prosperity	37.79	0.441	0.285	38.52	11.66
VGG16	**Ours**	9.38	0.194	0.285	**9.86**	1.81
ResNet18	PTB	11.16	0.667	0.077	11.91	11.91
ResNet18	SpikeX	11.93	1.060	0.614	13.61	6.72
ResNet18	LoAS	2.52	1.040	0.614	4.17	2.13
ResNet18	SATO	11.93	1.060	0.614	13.61	6.72
ResNet18	SpikeFlow	11.39	0.150	0.614	12.15	4.80
ResNet18	Prosperity	21.50	0.659	0.614	22.78	18.58
ResNet18	**Ours**	3.11	0.667	0.614	**4.39**	2.39

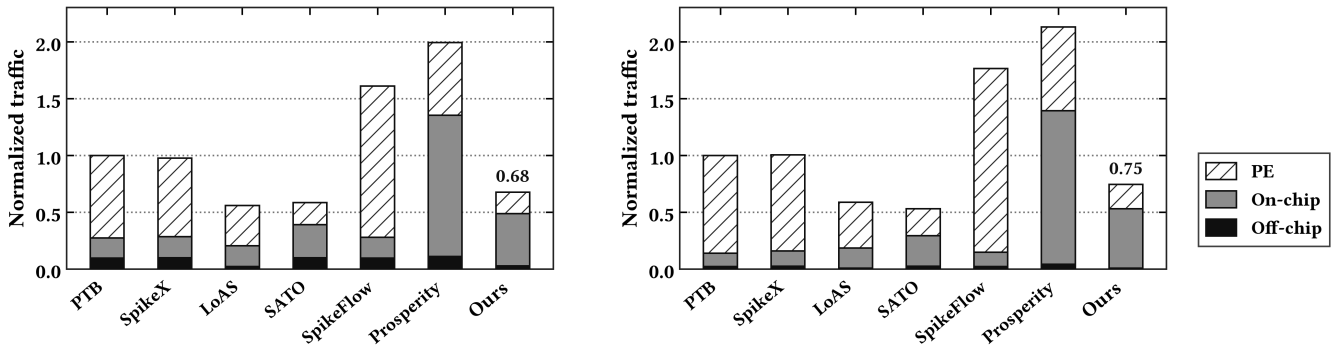


(a) VGG16 (b) ResNet18

Fig. 7. DRAM traffic per image (weights / input spikes / output spikes), normalized to PTB = 1, batch 1.

Traffic at three levels, batch 1. Off-chip: DRAM reads and writes. On-chip: every SRAM read and write (the counts that set the SRAM energy). PE: bits entering a PE from outside it (buffer, broadcast or a neighbouring PE). The levels include partial sums where a design moves them (ours: output-buffer read-modify-writes and the valid running sums passed down the columns; Prosperity: its prefix partial-sum buffer); the last two columns are weights and activations only.

Network	Design	Off-chip (MB)	On-chip (MB)	PE (MB)	Total (MB)	On-chip, wgt+act	PE, wgt+act
VGG16	PTB	33.9	61.3	250.8	346	61.3	250.8
VGG16	SpikeX	34.3	64.3	240.1	339	64.3	240.1
VGG16	LoAS	8.3	63.3	122.2	194	63.3	122.2
VGG16	SATO	34.3	101.4	67.1	203	101.4	67.1
VGG16	SpikeFlow	34.0	62.8	461.0	558	62.8	461.0
VGG16	Prosperity	38.5	430.6	220.8	690	95.9	52.3
VGG16	**Ours**	9.9	159.1	65.8	**235**	26.9	19.0
ResNet18	PTB	11.9	59.9	436.3	508	59.9	436.3
ResNet18	SpikeX	13.6	67.4	430.4	511	67.4	430.4
ResNet18	LoAS	4.2	89.9	206.1	300	89.9	206.1
ResNet18	SATO	13.6	135.6	121.9	271	135.6	121.9
ResNet18	SpikeFlow	12.1	63.5	821.8	898	63.5	821.8
ResNet18	Prosperity	22.8	686.1	375.6	1084	125.4	92.8
ResNet18	**Ours**	4.4	266.6	107.8	**379**	27.4	26.2



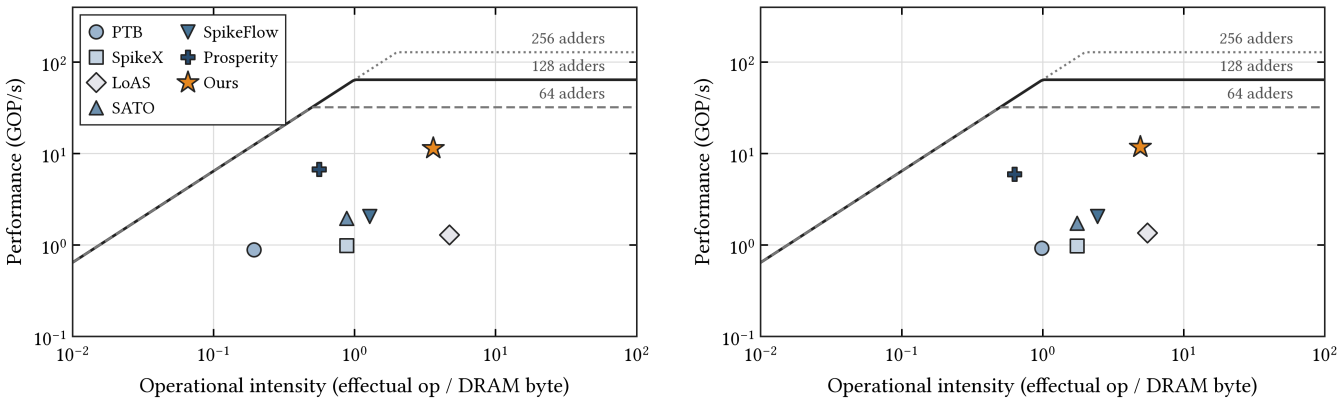
(a) VGG16 (b) ResNet18

Fig. 8. Traffic per image at three levels (off-chip / on-chip / PE), normalized to PTB = 1, batch 1.

Roofline (batch 16, 500 MHz, DRAM 64 GB/s). Work = effectual accumulations (non-zero weight × spike bit): VGG16 6.55 M, ResNet18 11.77 M per image; compute roof = adders × 500 MHz.

Network	Design	DRAM (MB/img)	OI (op/B)	Attained (GOP/s)	Compute roof (GOP/s)	Memory roof at OI (GOP/s)	Bound	% of roof
VGG16	PTB	33.86	0.193	0.88	64	12.38	memory	7%
VGG16	SpikeX	7.45	0.879	0.98	32	56.24	compute	3%
VGG16	LoAS	1.39	4.703	1.29	64	300.97	compute	2%
VGG16	SATO	7.45	0.879	1.95	64	56.24	memory	3%
VGG16	SpikeFlow	5.12	1.279	2.05	128	81.87	memory	3%

Network	Design	DRAM (MB/img)	OI (op/B)	Attained (GOP/s)	Compute roof (GOP/s)	Memory roof at OI (GOP/s)	Bound	% of roof
VGG16	Prosperity	11.66	0.562	6.72	64	35.94	memory	19%
VGG16	**Ours**	1.81	3.618	11.44	64	231.56	compute	18%
ResNet18	PTB	11.91	0.988	0.92	64	63.25	memory	1%
ResNet18	SpikeX	6.72	1.752	0.97	32	112.11	compute	3%
ResNet18	LoAS	2.13	5.532	1.36	64	354.07	compute	2%
ResNet18	SATO	6.72	1.752	1.73	64	112.11	compute	3%
ResNet18	SpikeFlow	4.80	2.452	2.04	128	156.95	compute	2%
ResNet18	Prosperity	18.58	0.634	5.91	64	40.55	memory	15%
ResNet18	**Ours**	2.39	4.933	11.80	64	315.69	compute	18%



(a) VGG16 (b) ResNet18

Fig. 9. Roofline, (a) VGG16, (b) ResNet18.

9. Dataflow ablation: WS vs IS vs OS

Same array (4 x 4 PEs, 8 adders per PE), same dual-side packing in all three, same workload (sparse90, batch 1). WS = our design (cycles RTL-verified); IS keeps packed pixel groups stationary and streams packed weight columns; OS gives each PE 4 filters x 4 pixels x 8 accumulators and processes one non-zero weight and one pixel group per cycle per input. IS and OS are analytical models with the same overheads. Energy: same core power for all, SRAM from CACTI per bit, same DRAM traffic.

Network	Dataflow	Cycles	Latency vs WS	Adder util.	Weight reads (MB)	Spike reads (MB)	Partial sums (MB)	Power (mW)	Energy (uJ)	EDP vs WS
VGG16	**WS (ours)**	364,930	1.00x	14.0%	4.3	10.6	126.6	2006	1464	1.00x
VGG16	IS	685,111	1.88x	7.5%	5.1	7.3	126.6	1206	1652	2.12x
VGG16	OS	1,151,984	3.16x	4.4%	5.5	25.5	0.3	739	1703	3.67x
ResNet18	**WS (ours)**	543,056	1.00x	16.9%	3.2	16.7	226.3	1130	1227	1.00x
ResNet18	IS	683,332	1.26x	13.5%	5.1	11.8	226.3	957	1307	1.34x
ResNet18	OS	1,902,135	3.50x	4.8%	4.3	43.2	0.6	427	1625	4.64x

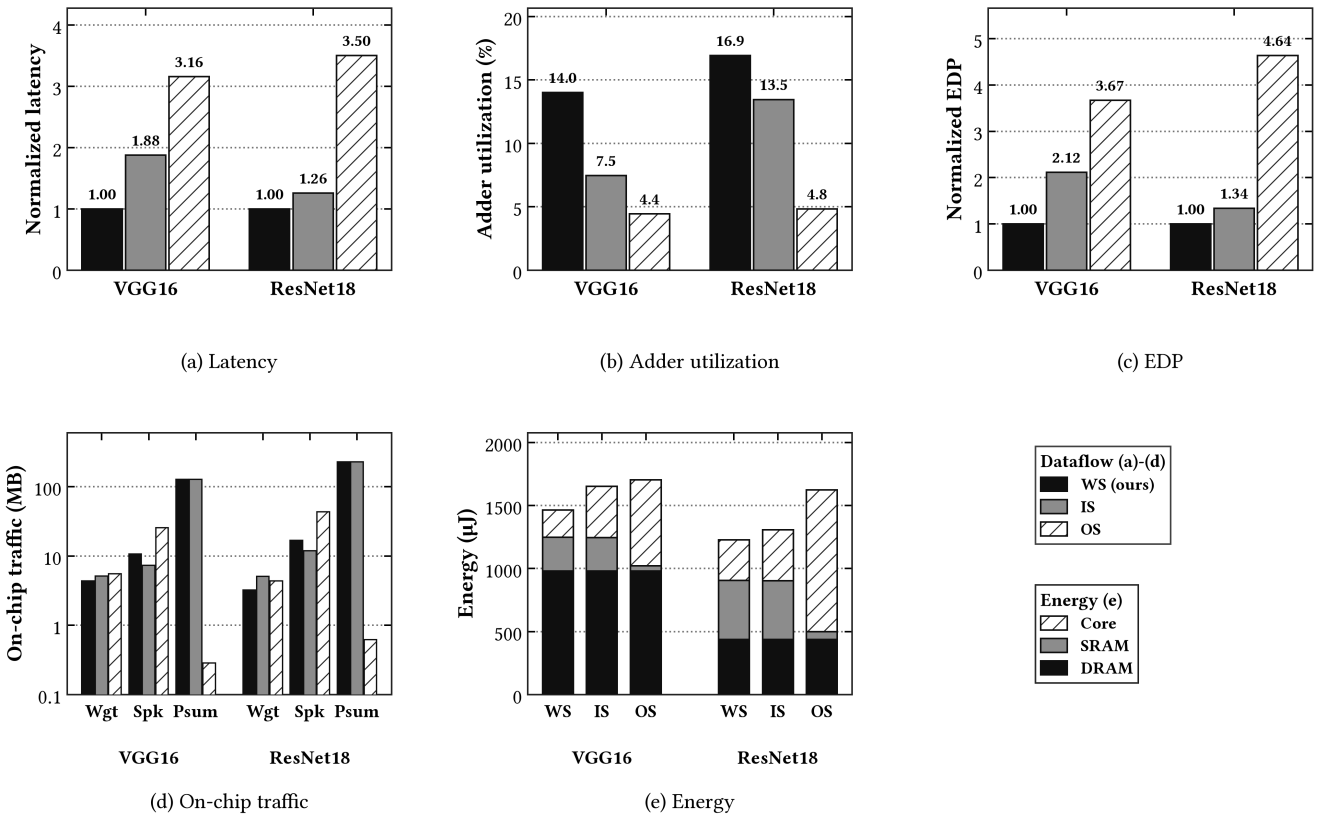


Fig. 10. Dataflow ablation: (a) latency, (b) adder utilization, (c) EDP, (d) on-chip traffic by type (log scale), (e) energy by component.

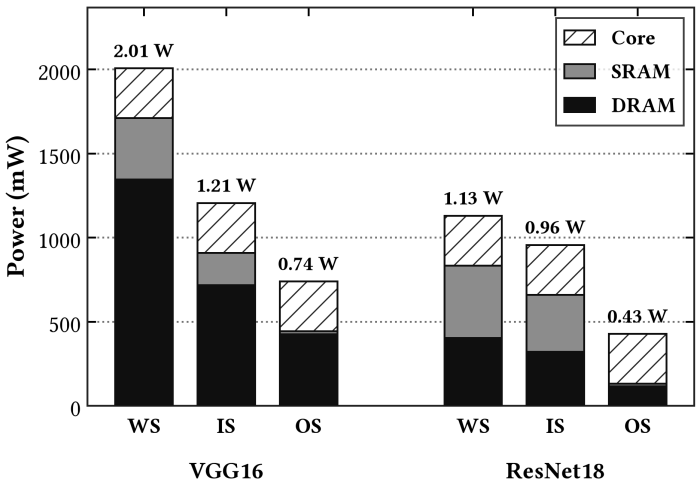


Fig. 11. Average power (energy / runtime) by component.

WS is the fastest (1.3-1.9x vs IS, 3.2-3.5x vs OS) and has the lowest EDP. Its cost is partial-sum traffic to the output buffer (about 90% of its on-chip traffic, 4.5-5x more than OS) and the highest average power, because the same DRAM and SRAM energy is spent in a shorter runtime.

10. Notes and limits

- Our timing is synthesis-level (no placement or wiring). The baselines' RTL was not timing-closed and is evaluated at the same 500 MHz; LoAS's TPPE does not close it (3.14 ns), so its GOP/s is optimistic.
- Prosperity's power is measured over a busy steady-state window (16 tiles, processor busy 87%); LoAS on the sparse90 stimulus. SATO's 54 KB buffer is assumed (not given in its paper). PTB has no cross-image batching (official model).
- Ungated flop clock power is most of every core's power (about 73% of ours); no design was clock-gated.
- Our DRAM count omits a per-tile column-count header (0.13 MB on VGG16, +1.4%).
- Not included because they predate the pipelined RTL and were not re-run: the design-space sweep, the bound audit, the weight-density 0.3 / 0.5 runs. Equal-PE (16 PE) comparisons are not included: every design runs at its paper's size.